

Rishi More

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Summary

ML systems engineer specializing in efficient and reliable LLM inference, from CUDA/Nsight kernel optimization and fused INT4 decoding to calibrated test-time compute and post-training evaluation. **Co-first author, ICML 2026 Main Conference**, on distribution-free risk control for cutting test-time reasoning compute under calibrated error guarantees.

Experience

JHU Data Science and AI Institute (DSAI)

May 2025 – June 2026

Graduate Research Assistant, LLM Inference Optimization

Baltimore, Maryland

- **Co-first author** of *Conformal Thinking* (**ICML 2026**, full paper): finite-sample calibration that halts reasoning early while holding a target error rate.
- Built a vLLM-based calibration/evaluation pipeline across **4** reasoning models and **4** benchmarks (**2.7K+** problems), maintained target error rates under dataset/length shift using finite-sample (UCB) calibration.
- Designed dual-threshold stopping rules, including a parametric lower threshold and ensembled halt policy, reducing reasoning tokens by **12%** at matched accuracy with reusable evaluation tooling.

JHU Center for Language and Speech Processing (CLSP)

January 2025 – December 2025

Graduate Researcher, Retrieval Systems

Baltimore, Maryland

- Built a temporal RAG pipeline over **10K+** time-stamped documents with time-decay weighting and metadata-scoped indexing to prioritize recent evidence over stale matches.
- Raised retrieval precision **23%** over a dense-embedding baseline by down-weighting outdated passages in fast-changing topics.
- Reduced stale and hallucination-prone retrievals **31%** via reference-free consistency scoring on retrieved context.

Mastek

November 2022 – July 2023

Machine Learning Engineer Intern

Mumbai, India

- Built and deployed an on-prem **edge** voice authentication system at **<100ms p99** latency and **96.6%** accuracy, using a lightweight Siamese mel-spectrogram model for retraining-free enrollment under edge-compute limits.
- Engineered a TensorFlow pipeline over **1M+** voice samples with spectral-gating denoising for robust feature extraction.
- Improved authentication reliability **20%** in high-noise production environments via preprocessing and feature-extraction upgrades.

Projects

Post-Training Failure Evaluation Harness | *Python, PyTorch, DPO, LLM-as-Judge* | GitHub

- Built a model-agnostic harness for **6** failure modes across a Qwen3-8B Base→SFT→DPO chain and a Qwen3-4B-Instruct comparator, evaluating **142** curated prompts.
- Implemented position-swapped pairwise judging, log-probability calibration, and per-example audit trails, finding that **10%** of DPO preference wins were factually degraded.
- Attributed regressions to individual optimization stages: DPO increased sycophancy from **34.1% to 44.9%**, while SFT raised harmful compliance from **0% to 12.5%**.

Scaffolded Multi-Agent Training for Language Agents | *Python, LLM Agents, LoRA, asyncio, BM25* | GitHub

- Built an asynchronous multi-agent training system in which a frozen **Qwen3-235B** caregiver taught LoRA-tuned **Qwen3-8B** agents through four-module memory and salience-gated BM25 retrieval.
- Across **160** training tasks and **3** seeds, caregiver conditions achieved **99.8–100%** task completion versus **81.5–84.4%** for Solo/Peer and required **26% fewer turns** on average (6.26 versus 8.44).
- On **40** held-out tasks, transfer accuracy remained between **68.2% and 70.8%** across all conditions, showing that the training advantage depended on caregiver scaffolding rather than independent transfer.

Decode Roofline: Kernel Profiling & Fused INT4 Dequant-GEMV | *CUDA C++, Nsight, vLLM* | GitHub

- Profiled batch-1 LLM decode (**Qwen2.5-1.5B**) at CUDA-kernel level, identifying **~81%** of kernel time in weight-streaming GEMVs pinned at **~95%** of the measured **~250 GB/s** memory roofline.
- Built correctness-gated fused **INT4** dequant + GEMV CUDA kernel hitting **86%** of peak bandwidth, reduced memory traffic by **~8.7×** versus a two-op baseline and improved batch-1 latency by **2.38×**.

Education

Johns Hopkins University

May 2026

M.S.E. in Computer Science

GPA: 3.9/4.0

- **Coursework:** Theory of Replicable ML, Human-in-the-Loop ML, Non-Stationary Environments.

University of Mumbai

July 2024

B.Tech. in Computer Science, Honors in AI/ML

CGPA: 9.85/10

Technical Skills

Languages: Python, C++, Java, C

ML/LLM Systems: PyTorch, TensorFlow, Hugging Face, Low-Rank Adaptation (LoRA), Direct Preference Optimization (DPO), retrieval-augmented generation (RAG), calibration, conformal/risk control, Scikit-learn

Inference & MLOps: vLLM, CUDA C++, Nsight Compute/Systems, Docker, Kubernetes, AWS, Apache Airflow, Apache Spark

Data: SQL, PostgreSQL, MongoDB, Pandas, NumPy